

Control Flow Statements

Simple if

if (condition) {
 statements
}

if-else

if (condition) {
 statements
}

else {
 statements;
}

1) get the mark from user. if the mark is greater 90 Print 'O Grade', or greater 80 Print 'A+ Grade', or greater 70 Print 'A Grade' else Print Fail.

```
int mark=scn.nextInt();
if(mark>90){
    System.out.println("O Grade");
}else{
    if(mark>80){
        System.out.println("A+ Grade");
    }else{
        if(mark>70){
            System.out.println("A Grade");
        }else{
            System.out.println("Fail");
        }
    }
}
```

"nested if-else".

else-if ladder

```
int mark=scn.nextInt();
if(mark>90){
    System.out.println("O Grade");
}else if(mark>80){
    System.out.println("A+ Grade");
}else if(mark>70){
    System.out.println("A Grade");
}else{
    System.out.println("Fail");
}
```

syntactical sugar of nested if-else.

Leap year (4, 100, 400)

24 hrs = 1 day

1 year = 365 days 5 hr 48 mins 48 secs $\approx$ 6 hrs

- ✓ 4 year (24 hrs)
- ✓ 8 year (24 hrs)
- ✓ 12 year (24 hrs)
- ✓ 40 year (24 hrs)
- ✓ 48 year (24 hrs)
- ✓ 80 year (24 hrs)
- ✓ 88 year (24 hrs)
- ✓ 96 year (24 hrs)
- ✗ 100 year (24 hrs - 18 hrs)
- ✓ 104 year (24 hrs)

100 (24-18)=6	500	✓ 1200
200 (24-18)=6	600	✓ 1600
300 (24-18)=6	700	✓ 2000
✓ 400 (24-18)=6	✓ 800	✓ 2400

24 hrs = 1 day

```
if(Y%4==0){  
    if(Y%100==0){  
        if(Y%400==0){  
            return true;  
        }else{  
            return false;  
        }  
    }else{  
        return true;  
    }  
}else{  
    return false;  
}
```

Looping $\rightarrow$ iterative or repetitive statement.

Entry control

$\rightarrow$ initially it checks the condition, if the condition is true then statement will execute.

$\rightarrow$ while, for

Exit control

$\rightarrow$ initially it executes the statement. then condition will be checked.

$\rightarrow$ do-while.

$\rightarrow$ if the given condition is true or false do-while loop will execute at least once.

Var dec;
while (condition) {
Statements;

Var updation, ← To make
3 our condition
as false.

```
int val=1;
while(val<=5){
    System.out.println("Rithisha");
    val++;
}
System.out.println(val);
```

for (var dec; condition; statement 1, statement 2, ..., var updation);
for (int i=1; i<=5; System.out.println("Prateeksha"), i++);

for (var dec; condition; var-updation) {
Statements;
}

```
for (int i=1; i<=5; i++){
    System.out.println("Prateeksha");
}
```

var dec;
do {
Statements;
} var-updation;
while (condition);

```
int i=1;
do{
    System.out.println("Subha");
    i++;
}while(i>=5);
```

break

it terminates the current block iteration.

```
for (int i=1; i<=10; i++){
    if (i==6) break;
    System.out.println(i);
}
System.out.println("Loop Ended");
```

1 2 3 4 5 Loop Ended.

Continue

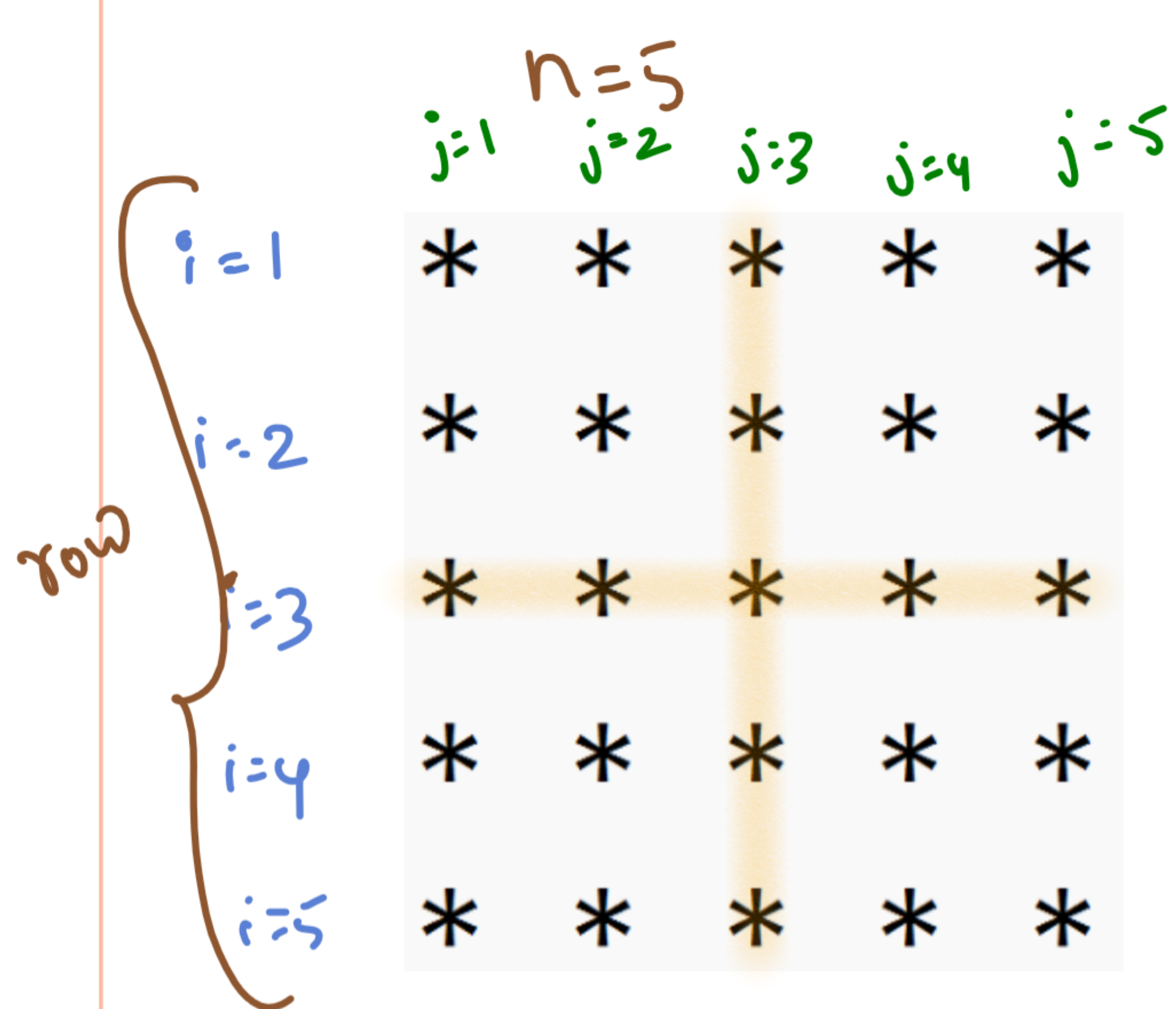
it skips the current iteration.

```
for (int i=1; i<=10; i++){
    if (i==6) continue;
    System.out.println(i);
}
System.out.println("Loop Ended");
```

1 2 3 4 5 7 8 9 10
Loop ended.

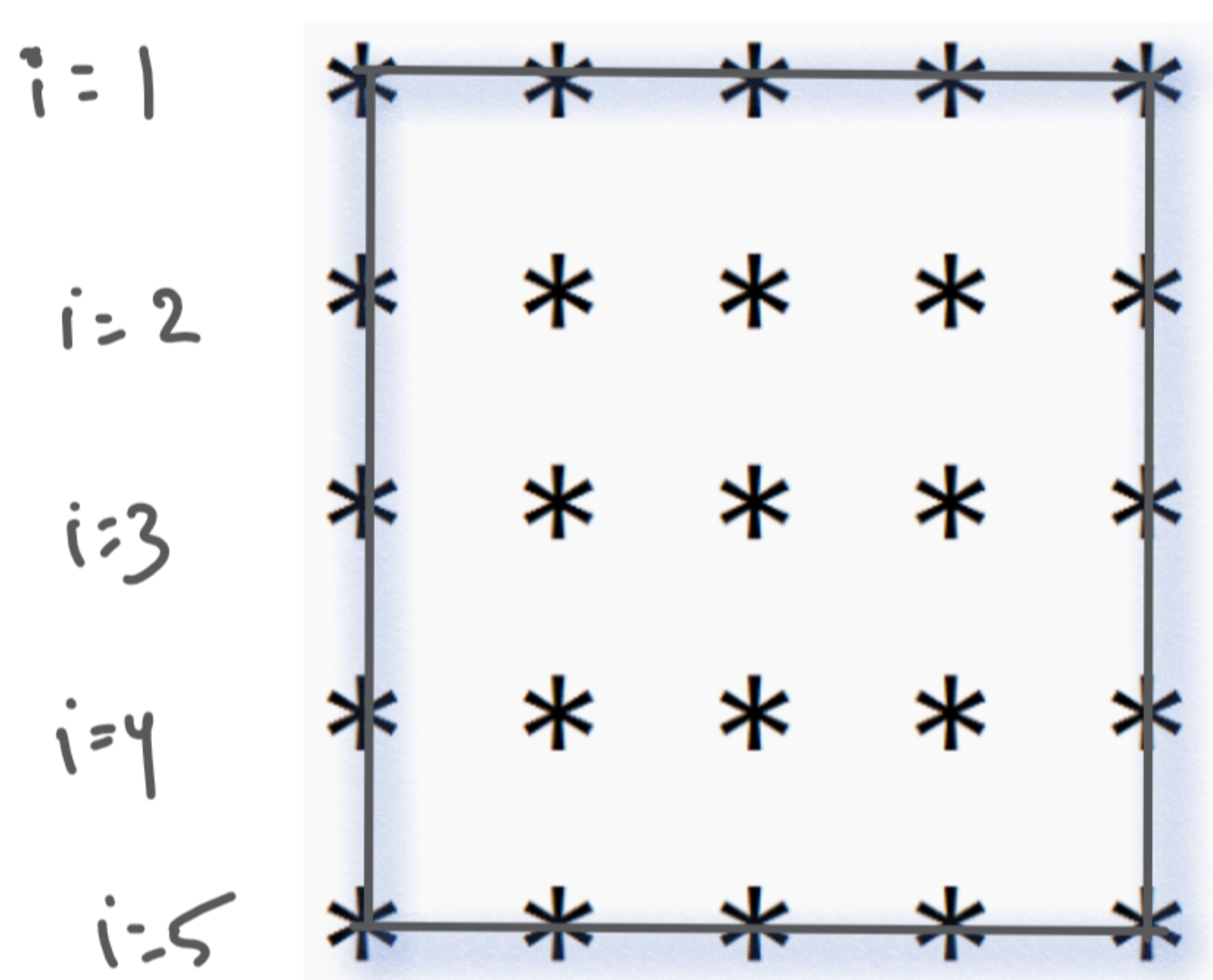
return

it terminates the entire Program.

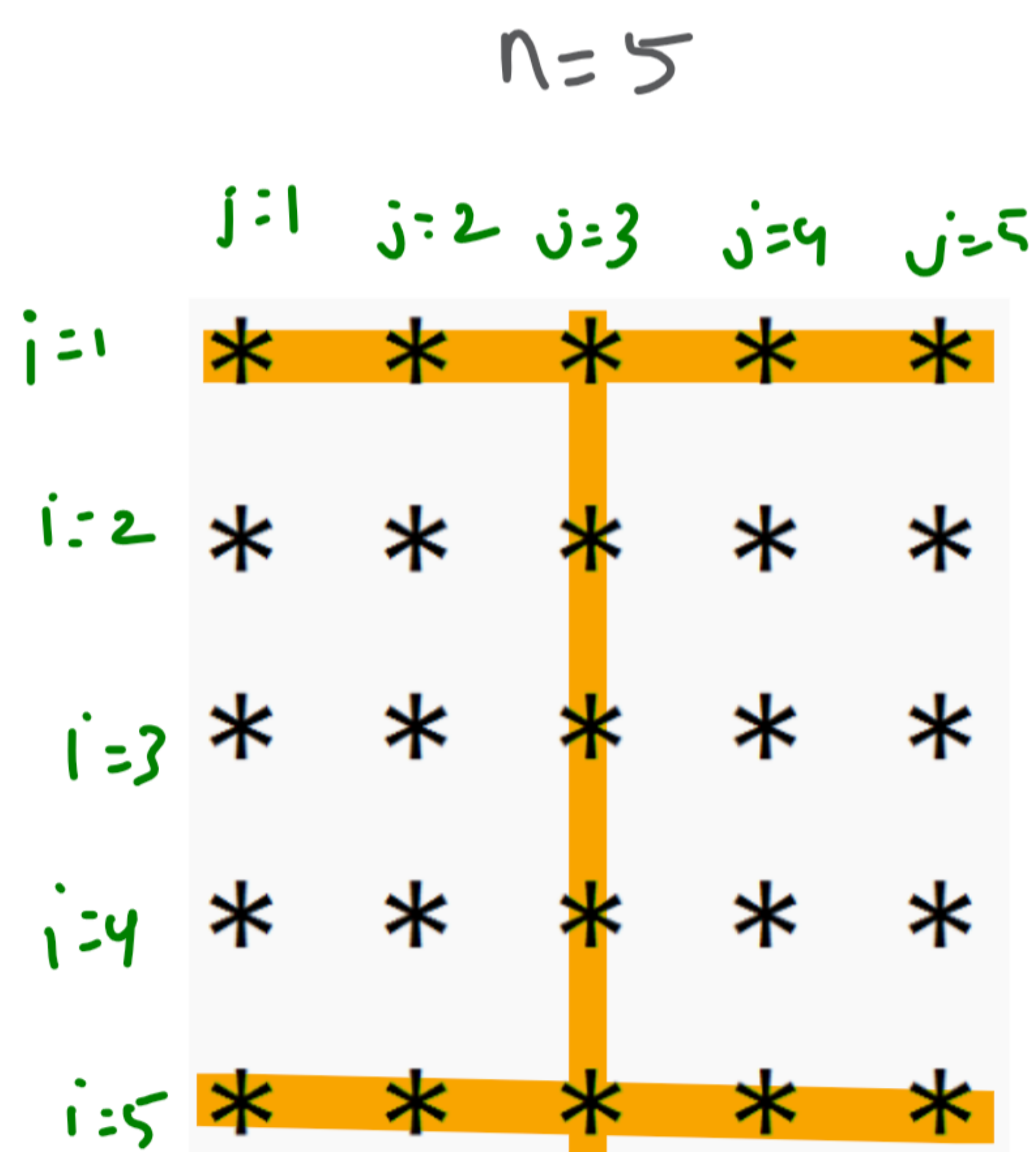
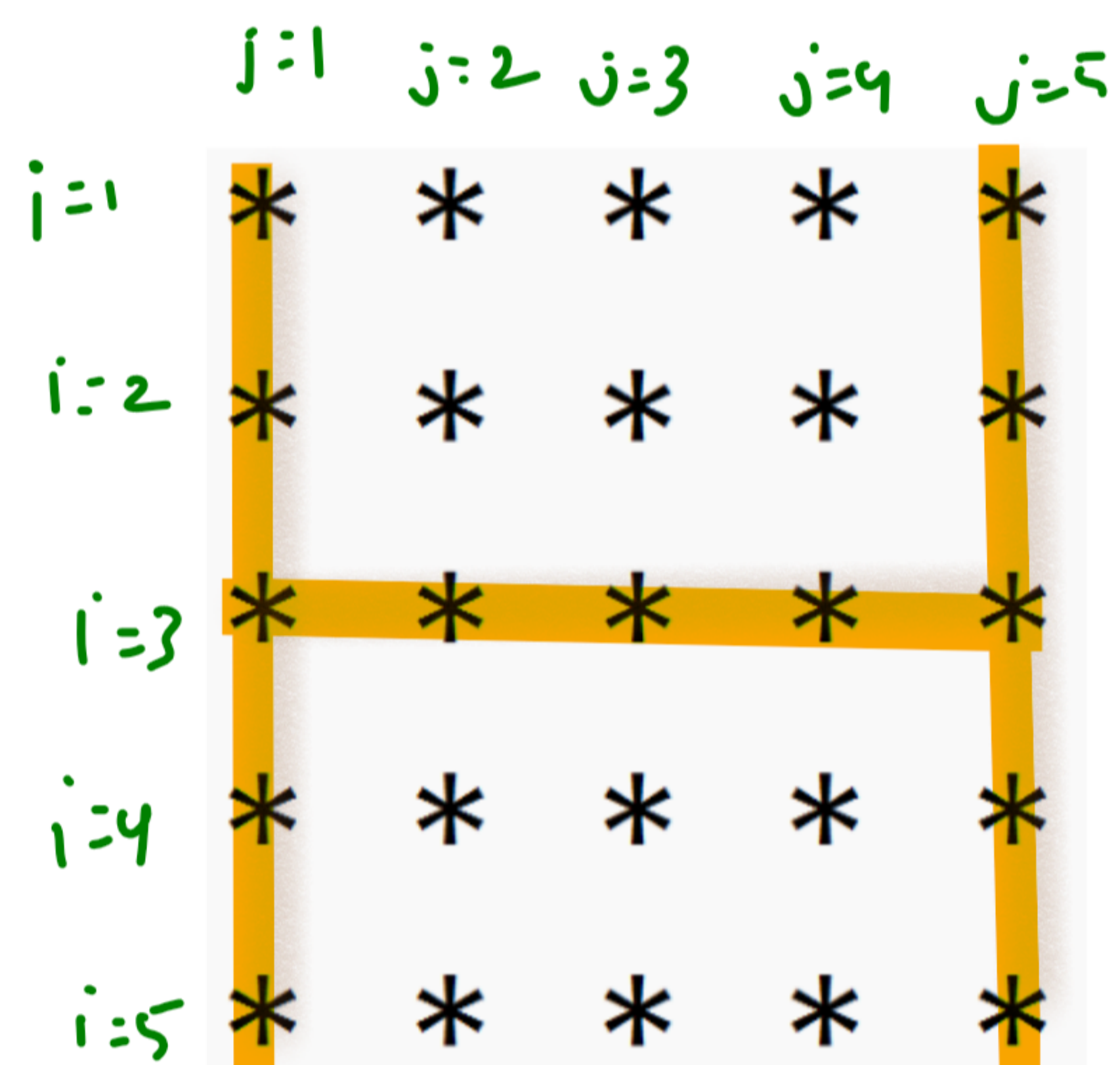


$$mid = \frac{n}{2} + 1$$

```
int n=scn.nextInt();
int mid=n/2+1;
for(int i=1;i<=n;i++){
    for(int j=1;j<=n;j++){
        if(i==mid||j==mid)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
    System.out.println();
}
```



```
if(i==1||i==n||j==1||j==n)
    System.out.print("* ");
else
    System.out.print("  ");
```

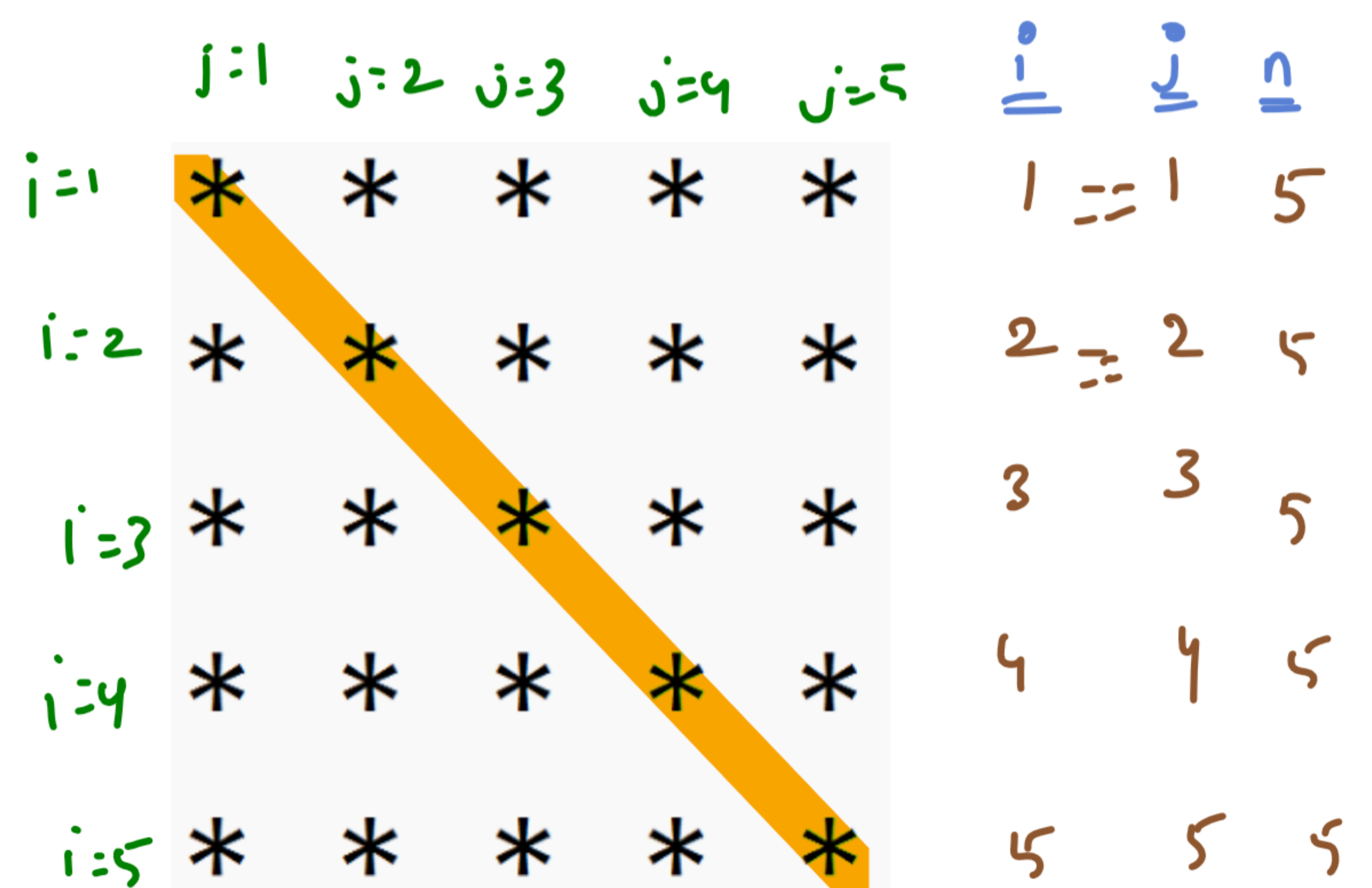


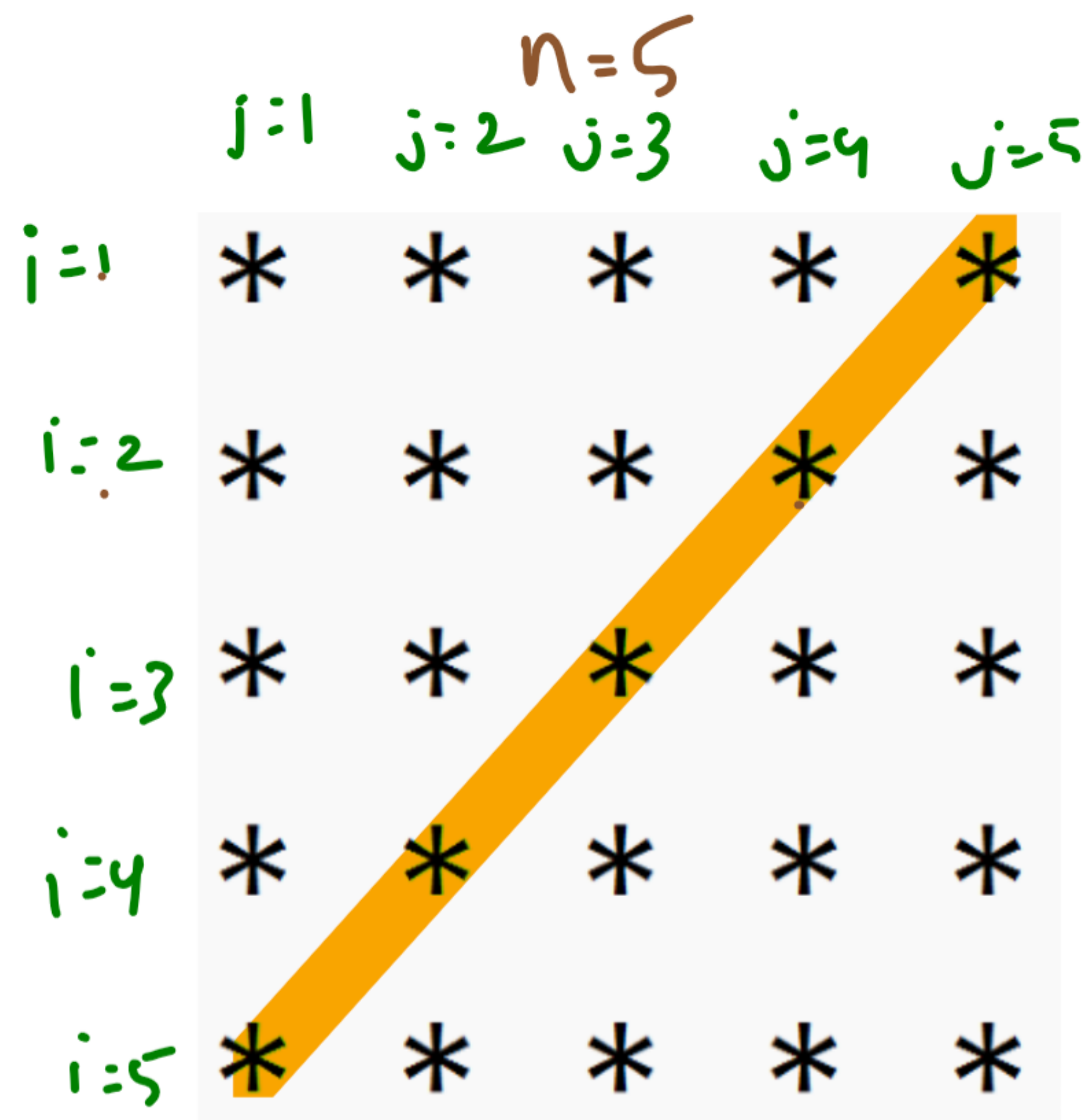
$n=5$

```
if(i==mid||j==n||j==1)
    System.out.print("* ");
else
    System.out.print("  ");
```

$n=5$

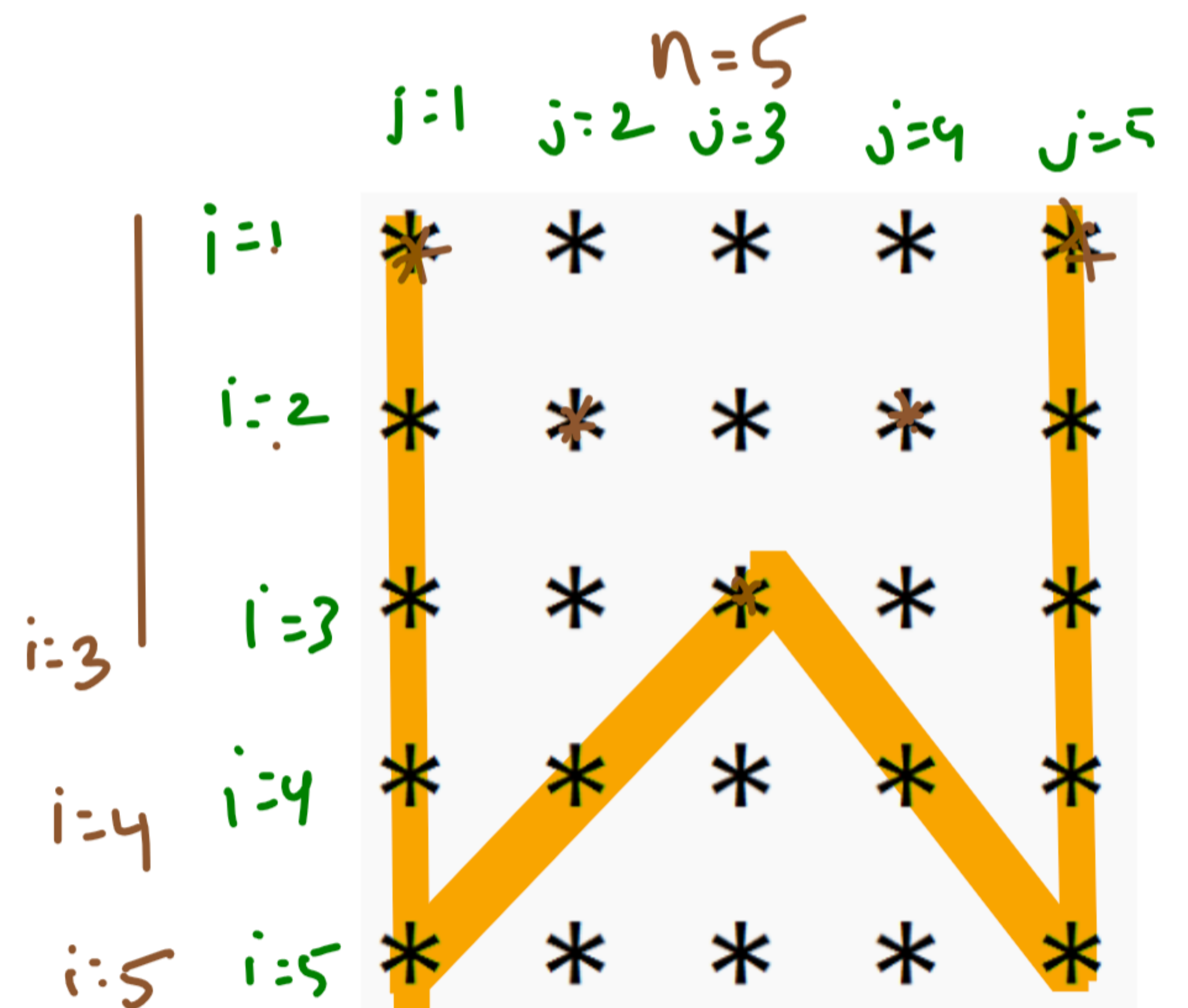
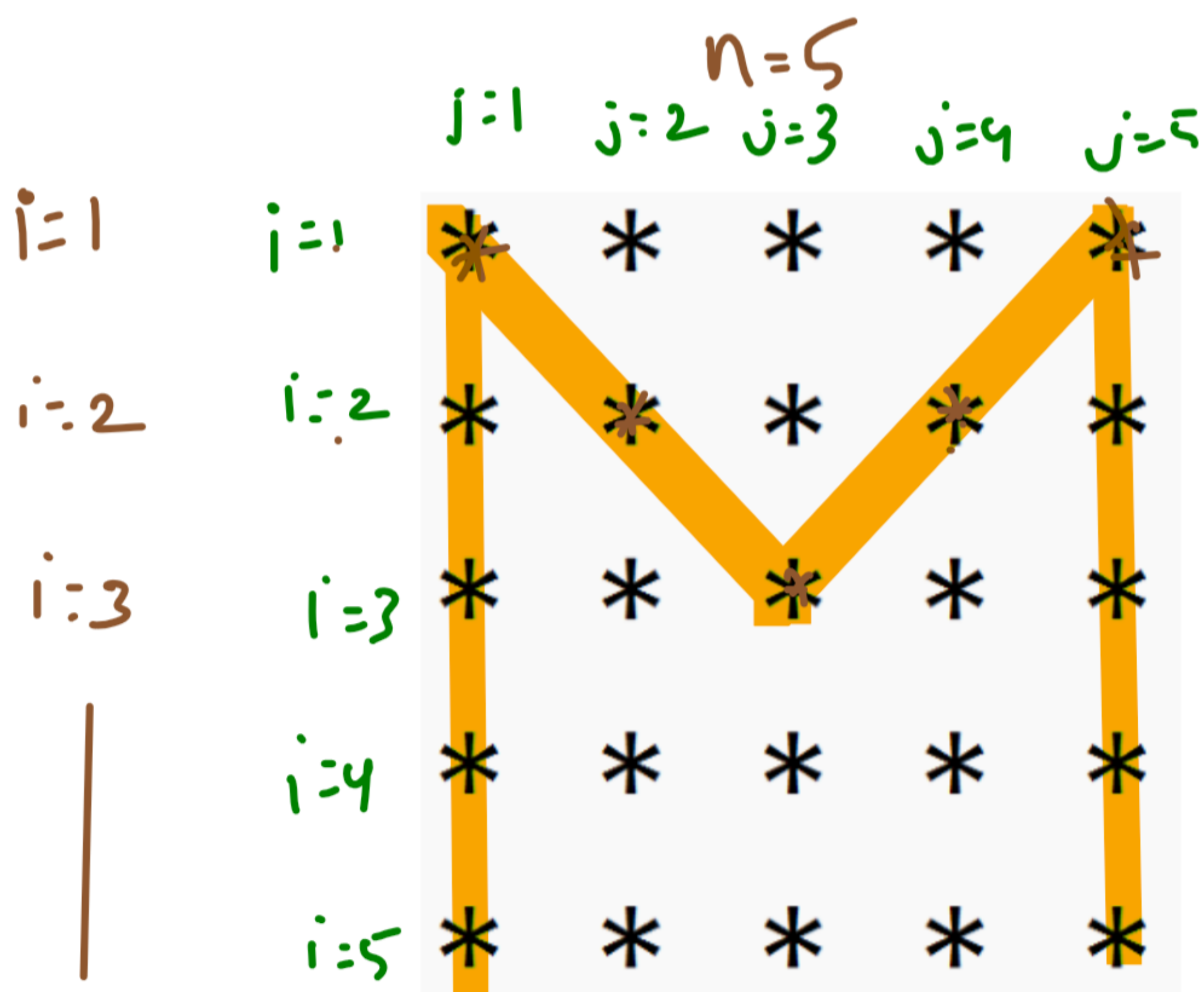
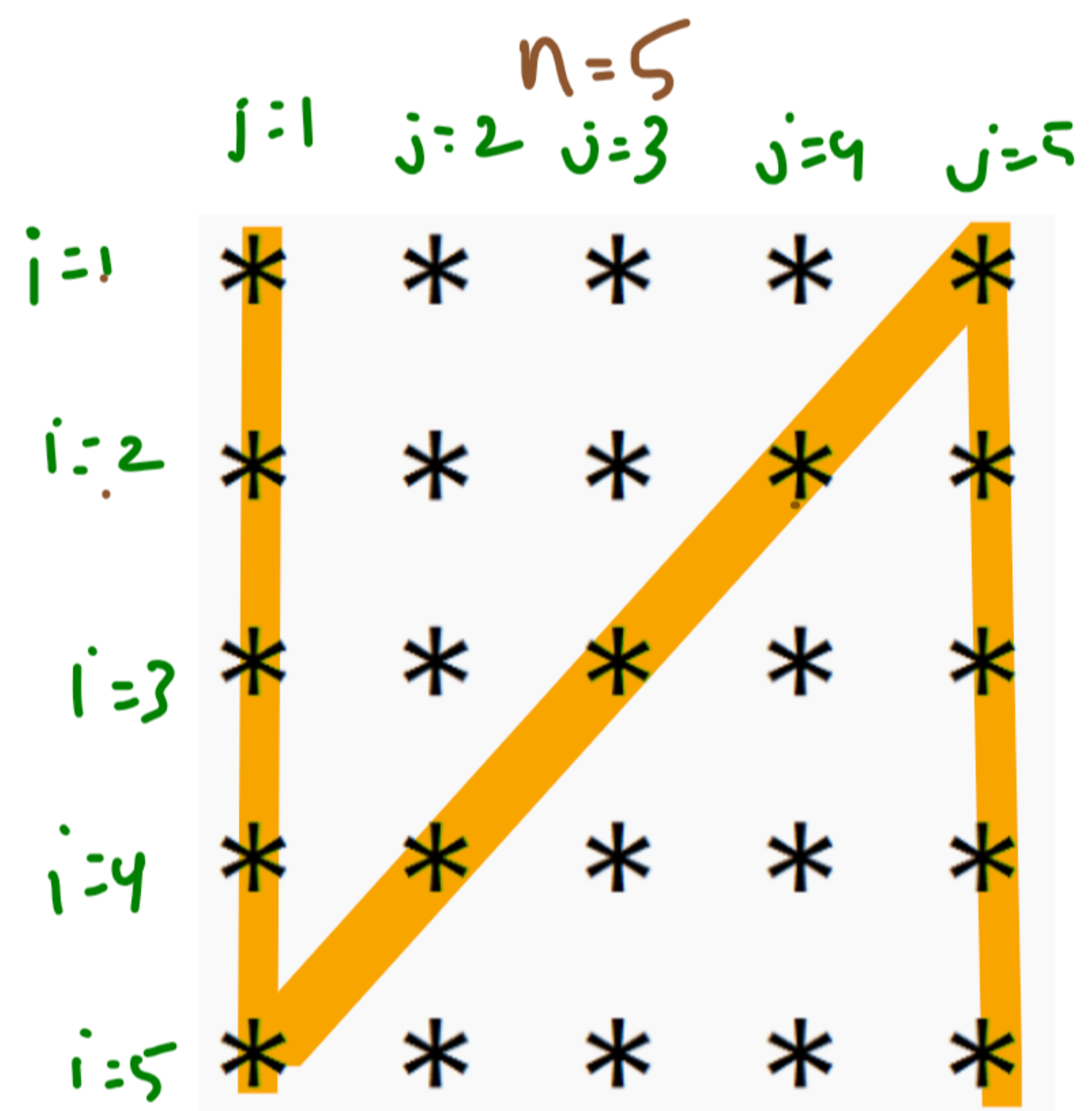
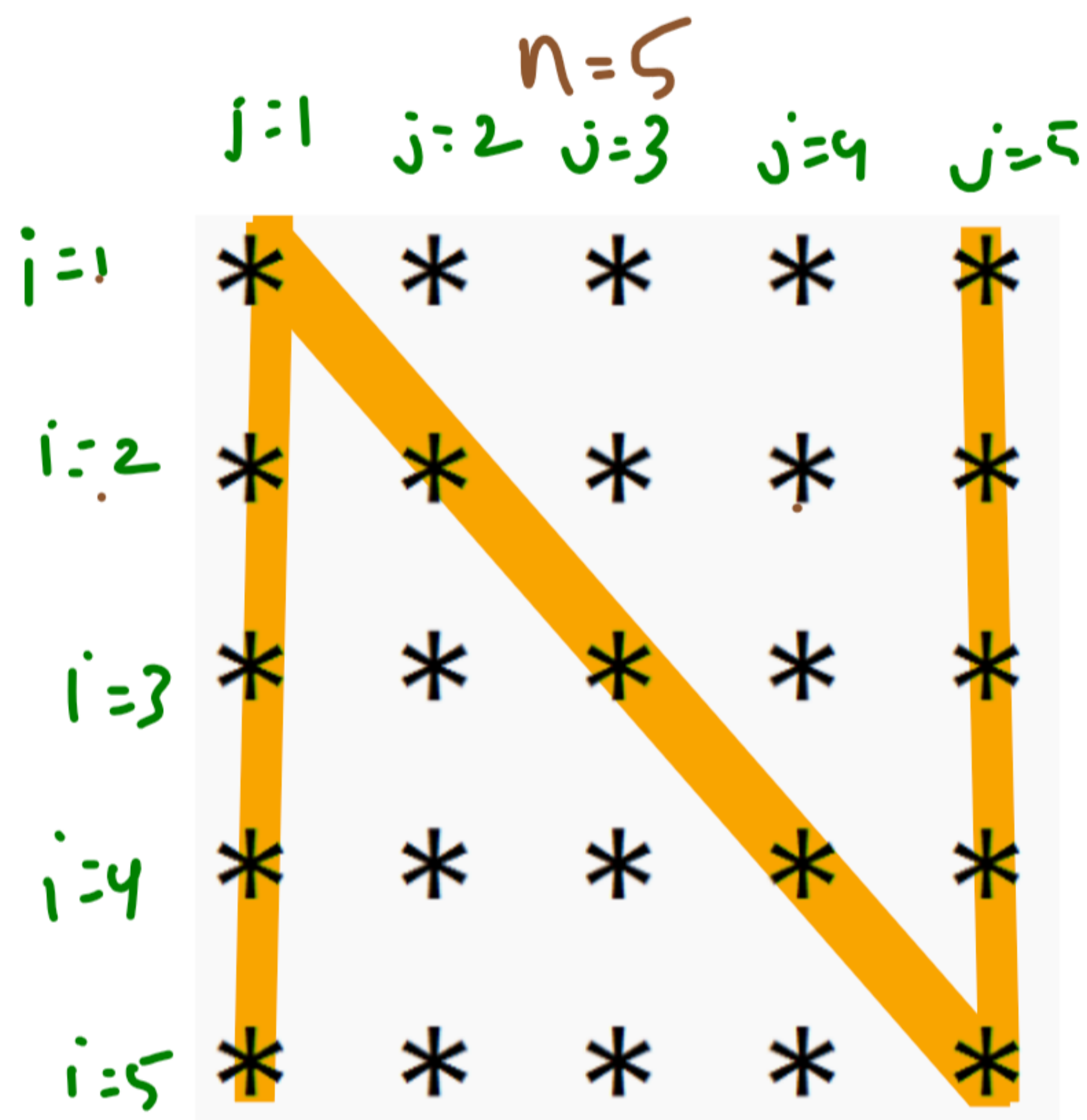
```
if(i==mid||j==n||j==1)
    System.out.print("* ");
else
    System.out.print("  ");
```





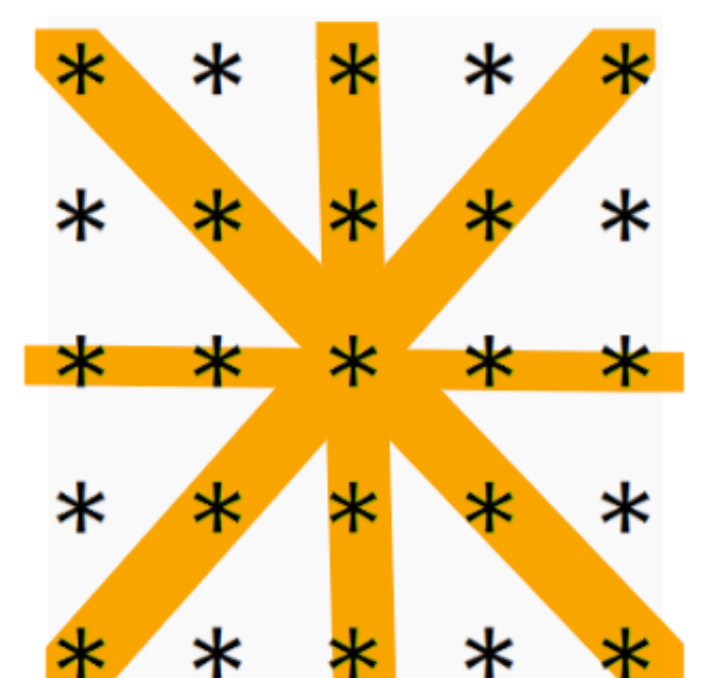
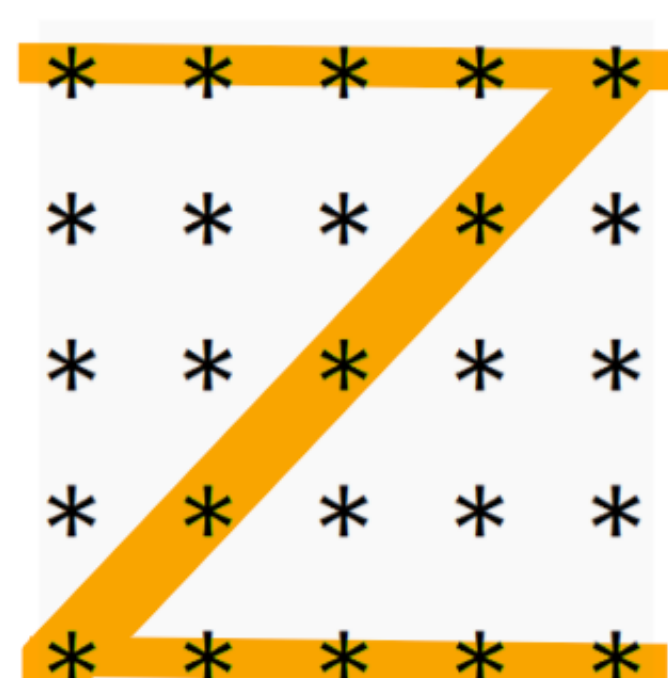
$$\begin{aligned} 1 + 5 &= 6 \quad (5+1) \quad 5 \\ 2 + 4 &= 6 \quad 5 \\ 3 + 3 &= 6 \quad 5 \\ 4 + 2 &= 6 \quad 5 \\ 5 + 1 &= 6 \quad 5 \end{aligned}$$

$$(i+j == n+1)$$



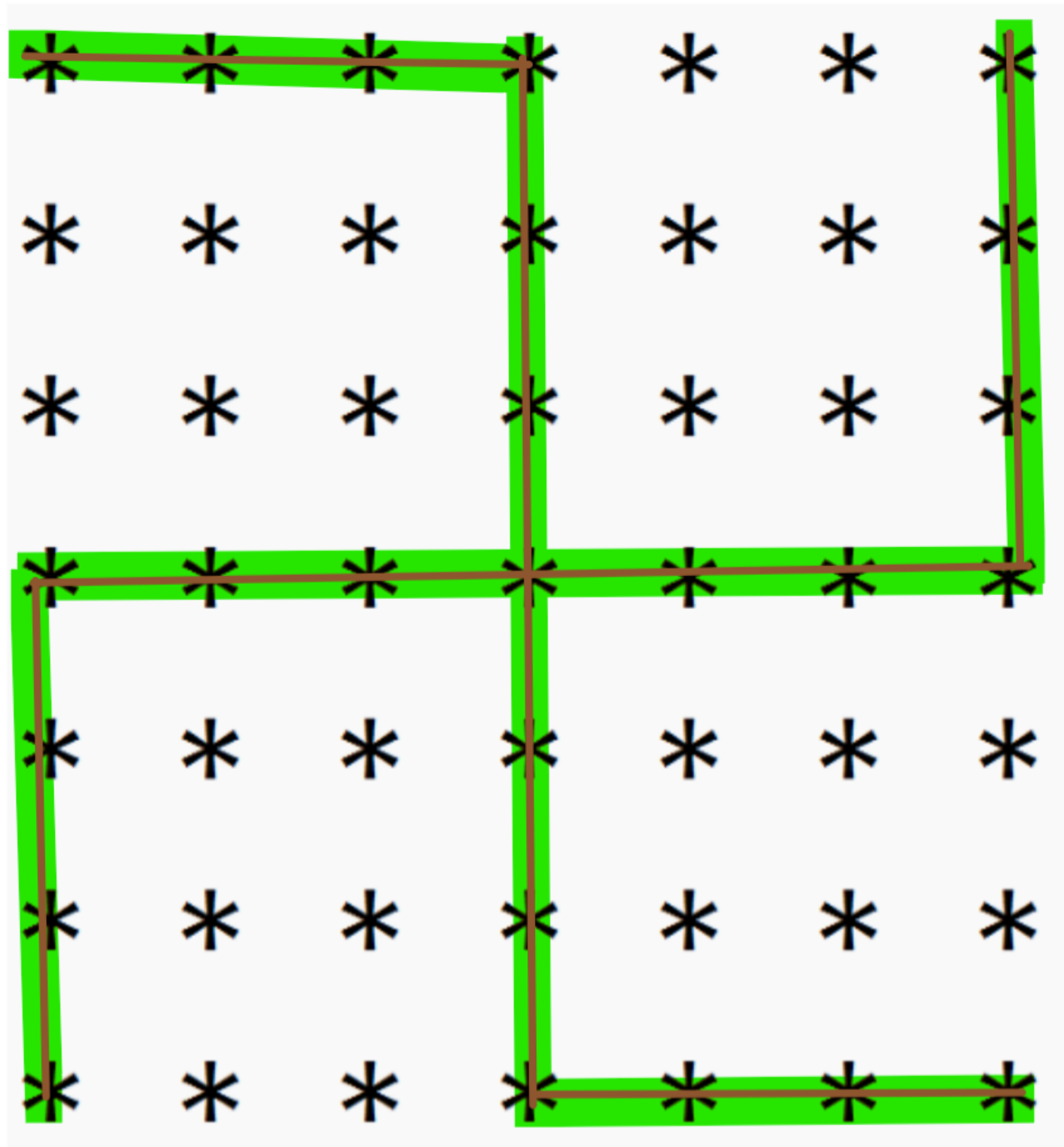
```
if((i>=mid&&(i==j||i+j==n+1))||(j==1)||(j==n))
    System.out.print("* ");
```

hands-on



$$i == 1 \text{ or } j \leq \text{mid}$$

$$j == n \text{ or } i \leq \text{mid}$$



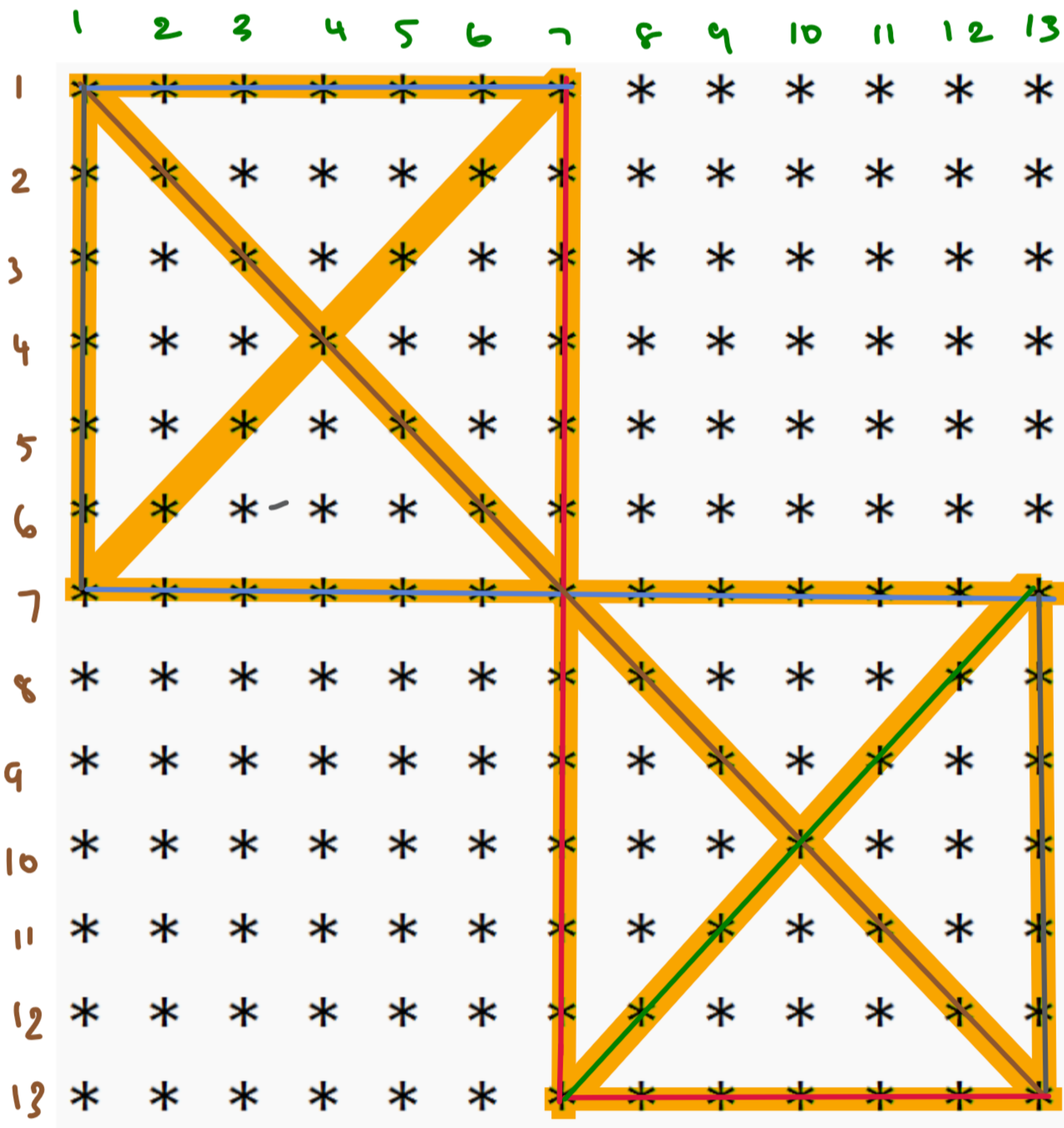
```
int n=scn.nextInt();
int mid=n/2+1;
for(int i=1;i<=n;i++){
    for(int j=1;j<=n;j++){
        if(i==mid || j==mid)
            System.out.print("* ");
        else if((i==1&&j<=mid)|| (i==n&&j>=mid))
            System.out.print("* ");
        else if((j==1&&i>=mid)|| (j==n&&i<=mid))
            System.out.print("* ");
        else
            System.out.print("  ");
    }
    System.out.println();
}
```

$$j == 1 \text{ or } i \geq \text{mid}$$

$$i == n \text{ or } j \geq \text{mid}$$

$$n=7$$

$$n=13$$



$$i == \text{mid}$$

$$j == \text{mid}$$

$$i == j$$

$$i == 1 \text{ or } j \leq \text{mid}$$

$$i == n \text{ or } j \geq \text{mid}$$

$$j == 1 \text{ or } i \leq \text{mid}$$

$$j == n \text{ or } i \geq \text{mid}$$

$$i+j == \text{mid}+1$$

$$i+j == n+\text{mid}$$

<u>i</u>	<u>j</u>	<u>mid</u>	<u>n</u>
1	7	7	13
2	6	7	13
3	5	7	13
4	4	7	13
5	3	7	13
6	2	7	13

$$i+j == \text{mid}+1$$

<u>i</u>	<u>j</u>	<u>mid</u>	<u>n</u>
7	13	7	13
8	12	7	13
9	11	7	13
10	10	7	13
11	9	7	13
12	8	7	13

$$2(n+\text{mid})$$

$$i+j == n+\text{mid}$$

Rules for Patterns

- 1) declare the credentials for 1st line.
str, sp
- 2) construct n of rows with given input.
- 3) make credentials loop with appropriate order.
- 4) update the credentials. for next line.

str=1 n=5

i=1 *
i=2 * * str=3
i=3 * * * str=4
i=4 * * * * str=5
i=5 * * * * * str=6
i=6 < 5

```
str=2
int str=1;//declare the credentials for 1st line
for(int i=1;i<=n;i++){//no of rows
    for(int j=1;j<=str;j++){//star loop
        System.out.print("* ");
    }
    str++;//update the credentials
    System.out.println();
}
```

n=5

```
* * * * *
* * * *
* * *
* *
*
```

```
int n=scn.nextInt();
int str=n;
for(int i=1;i<=n;i++){
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    str--;
    System.out.println();
}
```

SP=4
str=1

n=5

i=1 - - - - * , str=2, SP=3
i=2 - - - * * , str=3, SP=2
i=3 - - * * * , str=4, SP=1
i=4 - * * * * str=5, SP=0
i=5 * * * * * str=6, SP=-1
i=6

```
int sp=n-1, str=1;
for(int i=1;i<=n;i++){
    for(int j=1;j<=sp;j++){
        System.out.print(" ");
    }
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    str++;
    sp--;
    System.out.println();
}
```

n=5

```
* * * * *
  * * * *
    * * *
      * *
        *
```

Try hands-on

```
int n=scn.nextInt();
int str=1,sp=2*n-3;
for(int i=1;i<=n;i++){
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    for(int j=1;j<=sp;j++){
        System.out.print(" ");
    }
    if(i==n) str--;
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    str++;
    sp-=2;
    System.out.println();
}
```

(2n-3)
n sp
3 3
5 7
7 11

n=5

```
* - - - - *
* * - - - * *
* * * - - * * *
* * * * - * * * *
* * * * * * * *
```

str=2, sp=5
str=3, sp=3
str=4, sp=1
str=5, sp=-1

n=7

sp=0, str=7

```
i=1 * * * * *
i=2   * * * *
i=3     * * *
i=4       *
i=5        * * *
i=6   * * * * *
i=7 * * * * * *
```

sp=1, str=5
sp=2, str=3
sp=3, str=1
sp=2, str=3
sp=1, str=5
sp=0, str=7

if(i<n/2)

```
int n=scn.nextInt();
int str=n,sp=0;
for(int i=1;i<=n;i++){
    for(int j=1;j<=sp;j++){
        System.out.print(" ");
    }
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    if(i<=n/2){//upper bound
        sp++;
        str-=2;
    }else{//lower bound
        sp--;
        str+=2;
    }
    System.out.println();
}
```

n=7

```
      *
    * * *
  * * * * *
* * * * * * *
  * * * * *
    * * *
      *
```

```
int str=1,sp=n/2;
for(int i=1;i<=n;i++){
    for(int j=1;j<=sp;j++){
        System.out.print(" ");
    }
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    if(i<=n/2){
        sp--;
        str+=2;
    }
    else{
        sp++;
        str-=2;
    }
    System.out.println();
}
```

n=5

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
int str=n/2+1,sp=1;
for(int i=1;i<=n;i++){
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    for(int j=1;j<=sp;j++){
        System.out.print(" ");
    }
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    if(i<=n/2){
        sp+=2;
        str--;
    }
    else{
        sp-=2;
        str++;
    }
    System.out.println();
}
```

Space loop

n=7

Str loop

```
i=1 *
i=2 * *
i=3 * * *
i=4 * * * * *
i=5 * * *
i=6 * *
i=7 *
```

```
int str=1,sp=n/2+1;
for(int i=1;i<=n;i++){
    for(int j=1;j<=sp;j++){
        if(i==n/2+1)
            System.out.print("* ");
        else
            System.out.print(" ");
    }
    for(int j=1;j<=str;j++){
        System.out.print("* ");
    }
    if(i<=n/2){
        str++;
    }
    else{
        str--;
    }
    System.out.println();
}
```